

Descriptive Statistics Project

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Project:

Descriptive Statistics of the Number of Capital Prisoners in 2017 in eleven western U.S. states.

Retrieved From: 'MyLab Statistics' Question 10, Section 3.4.58: Comparing Measures of Center.

Objectives:

- (1.) Data Presentation: I will present my dataset using dotplot and describe it.
- (2.) Measures of Center: I will calculate and interpret the mean, median, mode, and midrange of the dataset.
- 3.) Measures of Spread: I will calculate and interpret the range, variance, standard deviation, and interquartile range of the dataset.

(4.) Measures of Position: I will calculate and interpret the Five-Number Summary of the dataset.

Dataset

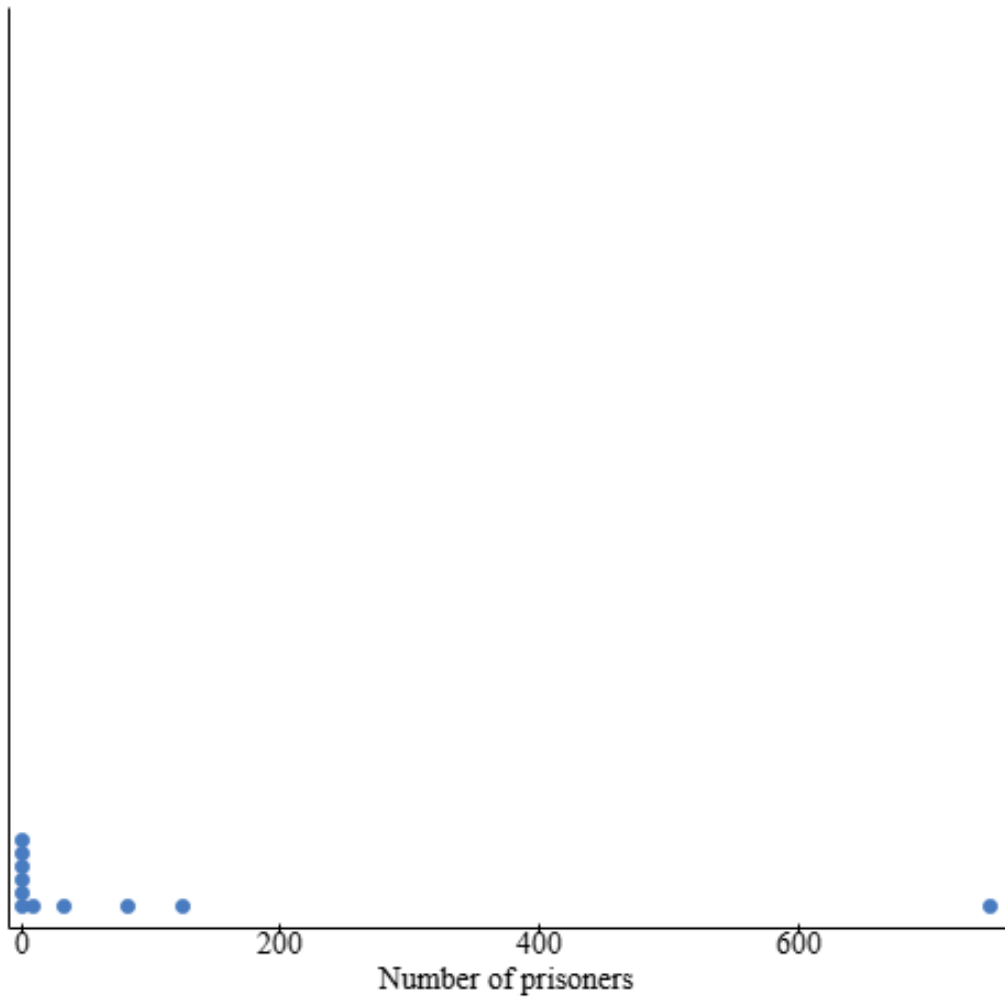
State	Capital Prisoners
Colorado	3
Arizona	125
Utah	9
New Mexico	2
California	746
Wyoming	1
Nevada	82
Montana	2
Oregon	33
Alaska	0
Hawaii	0

Data Values:

3,125, 9, 2, 746, 1, 82, 2, 33, 0, 0.

Description: The Number of Capital Prisoners in 2017 in eleven western U.S. states.

Presentation: I will use a Dotplot.



(1) This dataset is skewed right.

(2) The state of California has an unusually high number of capital prisoners.

This is seen as a potential outlier.

Analysis:

Measures of Center:

Let x be the number of Capital Prisoners in different U.S. states.

$$\Sigma x = 3 + 125 + 9 + 2 + 746 + 1 + 82 + 2 + 33 + 0 + 0 = 1,003$$

$$n = 11$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{1003}{11} = 91.18181818$$

Mean: The average number of capital prisoners in eleven western U.S. states in 2017 is approximately 92 capital prisoners.

Dataset in ascending order is:

0, 0, 1, 2, 2, 3, 9, 33, 82, 125, 746

$$\text{Median} = Q_2 = 3$$

Median: When the number of capital prisoners are sorted in ascending order, the median number of capital prisoners is 3 capital prisoners.

Mode: Two states have 0 capital prisons, which makes the mode 0 capital prisoners.

The minimum number of capital prisoners is 0.

The maximum number of capital prisons is 746.

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{0 + 746}{2} = 373$$

Midrange: The average of the minimum number and the maximum number of capital prisoners in eleven western U.S. states in 2017 is 373 capital prisoners.

Measures of Spread:

Let us calculate the range.

$$\text{Range} = \text{maximum} - \text{minimum}$$

$$\text{Range} = 746 - 0 = 746$$

Range: The difference between the maximum number and the minimum number of capital prisoners in eleven western U.S. states in 2017 is 746 capital prisoners.

Let us calculate the variance and the standard deviation.

x	$x - \bar{x}$	$(x - \bar{x})^2$
3	-88.1818181818	7776.0330578512
125	33.8181818181	1143.6694214876
9	-82.1818181818	6753.8512396694
2	-89.1818181818	7953.3966942148
746	654.8181818181	428786.85123967
1	-90.1818181818	8132.7603305785
82	-9.1818181818	84.3057851239
2	-89.1818181818	7953.3966942148
33	-58.1818181818	3385.1239669421
0	-91.1818181818	8314.1239669421
0	-91.1818181818	8314.1239669421

$$\Sigma(x - \bar{x})^2 = 488597.6364$$

$$\text{Variance, } s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{488597.6364}{11 - 1} = 48859.76364$$

$$\text{Standard Deviation, } s = \sqrt{48859.76364} = 221.0424476$$

Standard Deviation: This value of standard deviation tells us that there is a lot of spread in the number of capital prisoners in eleven western states in 2017.

Next is the Interquartile range.

Interquartile Range:

Dataset in ascending order is:

0, 0, 1, 2, 2, 3, 9, 33, 82, 125, 746

$$n = 11$$

$$\frac{1}{4} * 11 = 2.75$$

Rounded up to 3

$$3rd\ value = 1$$

$$Lower\ Quartile, Q_1 = 3rd\ value = 1$$

$$\frac{3}{4} * 11 = 8.25$$

Rounded up to 9

$$9th\ value = 82$$

$$Upper\ Quartile, Q_3 = 9th\ value = 82$$

Interquartile Range, $IQR = Q_3 - Q_1$

$$IQR = 82 - 1 = 81$$

Interquartile Range: The variation in 50% of the number of capital prisoners for eleven western U.S. states in 2017 is 81 capital prisoners.

Measures of Position:

The five-number summary of the dataset and their interpretations are:

$$\text{Minimum} = 0$$

The minimum number of capital prisoners is 0. Two U.S. states: Alaska and Hawaii account for that number.

$$Q_1 = 1$$

25% of the number of capital prisoners is below 1.

$$Q_2 = 3$$

50% of the number of capital prisoners is below 3.

$$Q_3 = 82$$

75% of the number of capital prisoners is below 82.

Maximum = 749

The maximum number of capital prisoners is 749. One U.S. state: California accounts for that number.

Works Cited (MLA 9)

Chukwuemeka, Samuel. "Descriptive Statistics Projects." *Descriptive Statistics*, 2024, conferencepresentations.appspot.com/Projects/Statistics/DescriptiveStatistics.html.

Comparing Measures of Center, Pearson, mylab.pearson.com/Student/PlayerHomework.aspx?homeworkId=688347288&questionId=11&flushed=true&cId=8105272&back=https%3A%2F%2Fmylab.pearson.com%2FStudent%2FDoAssignments.aspx%3Fcanvas_user_id&&canvas_enrollment_state=active&canvas_person_timezone=America%2FNew_York&context_parent_id=8046effc99bb470ab17693651a7c70f3&canvas_api_domain=learn.vccs.edu&canvas_user_login_id=hb96288&canvas_course_startat=&canvas_course_id=675765&canvas_workflow_state=available&orig_context_id=c037c6930a7db8f37229602de993b23c4aad773a&canvas_term_startat=%24Canvas.term.startAt&canvas_term_name=BRCC+Spring+2025&canvas_course_previous_ids=&tool_consumer_info_product_family_code=canvas&tool_consumer_info_version=cloud&lis_course_offering_sourceid=BR291.MTH.155.03.SP25&lis_person_sourcedid=7941636&view=all. Accessed 9 Apr. 2025.